Introduction to Computational Chemistry

What is Computational Chemistry?

- Computational chemistry is a branch of chemistry that uses the results of theoretical chemistry incorporated into efficient computer programs to calculate the structures and properties of molecules and solids, applying these programs to real chemical problems.

- Chemistry in the computer instead of in the laboratory

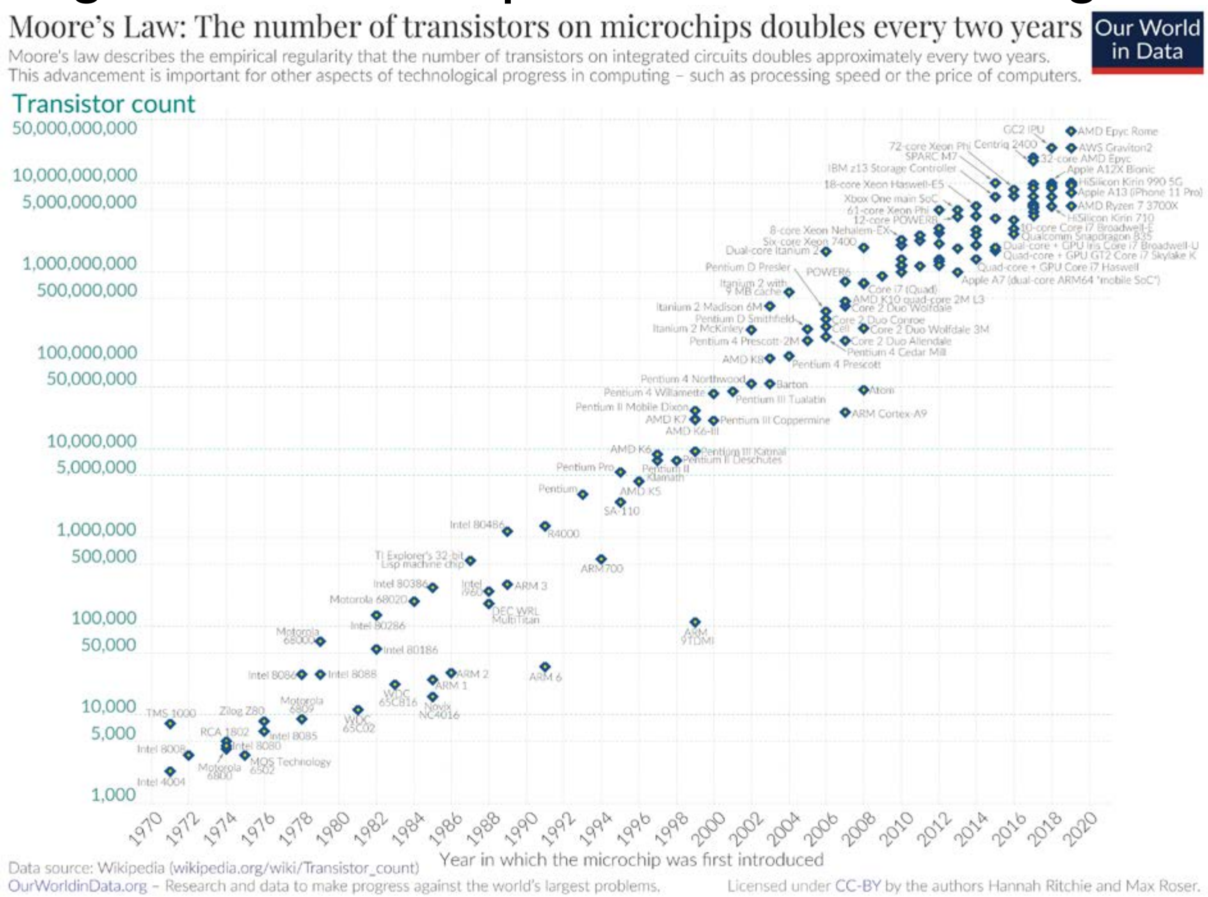
- Computational chemists, in contrast, may simply apply existing computer programs and methodologies to specific chemical questions.

- Use computer calculations to predict the structures, reactivities and other properties of molecules

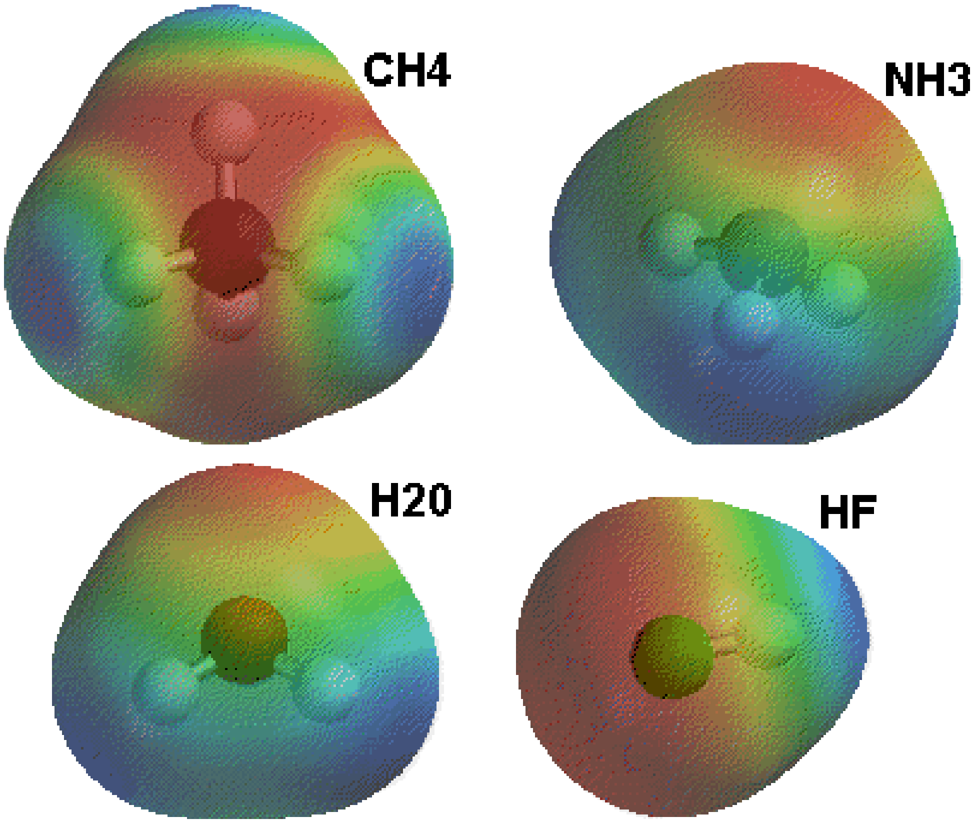
What is Computational Chemistry?

- Computational chemistry has become widely used because of
- -dramatic increase in computer speed
- -design of efficient quantum chemical algorithms

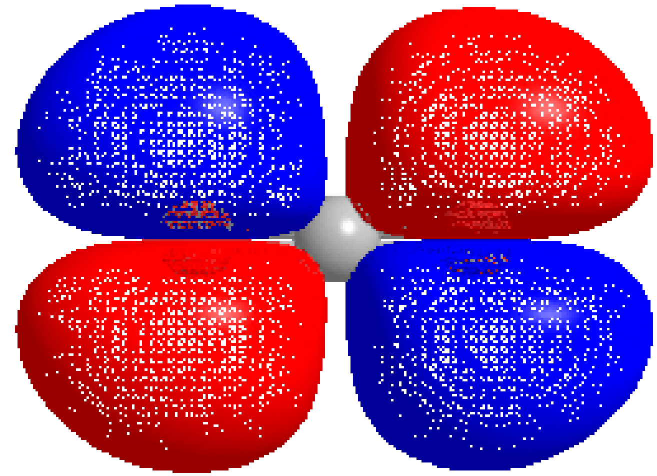
Moore's Law



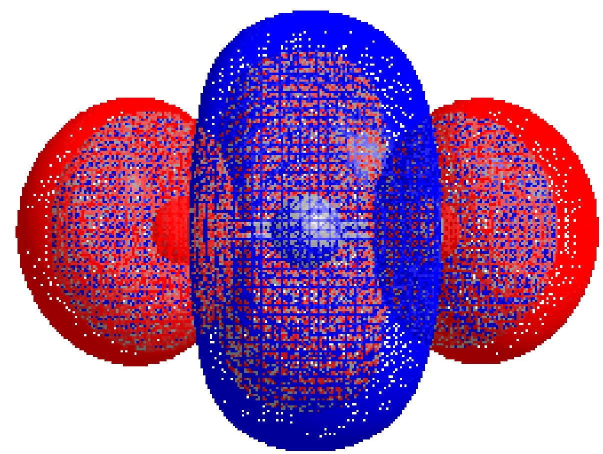
Motivation



Electropotential map



3D model of highest occupied molecular orbital in $\underline{CO_2}$



3D model of lowest unoccupied molecular orbital in CO_2

Why do Chemistry on a Computer?

How will you visit to Chittagong?

Why do Chemistry on a Computer?

Chemistry was done without computers till 1960. What is the need now?

- How will you visit to Chittagong? By walk!!
- You can calculate a 12 X 12 determinant by hand !
- Chemistry involves molecular structures and changes in molecular structures in reactions.
- These structures can only be computed using programs that involve matrices, determinants, integration, roots of equations, differential equations, interpolation, ...
- Bulk properties (liquids, solutions, ..) need statistical averages

Why do Chemistry on a Computer?

- Calculations are easy to perform whereas experiments are difficult
- Calculations are safe whereas many experiments are dangerous
- Calculations are becoming less costly while experiments are becoming more expensive
- Calculations can be performed on any chemical system, whereas experiments are relatively limited
- Calculations give direct information whereas there is often uncertain in interpreting experimental observation
- Calculations give fundamental information about isolated molecules without the complicating solvent effects
- It also helps chemists make predictions before running the actual experiments so that they can be better prepared for making observations.

Computational Chemistry

At present the use of computers to aid chemical inquiry, including, but not limited to:

- **Semi-Empirical Molecular Orbital Theory (Quantum Mechanics)**
- **Ab Initio Molecular Orbital Theory (Quantum Mechanics)**
- **Density Function Theory (Quantum Mechanics)**
- **Molecular Mechanics (Classical Newtonian Physics)**
- **Molecular Dynamics (Classical Newtonian Physics)**

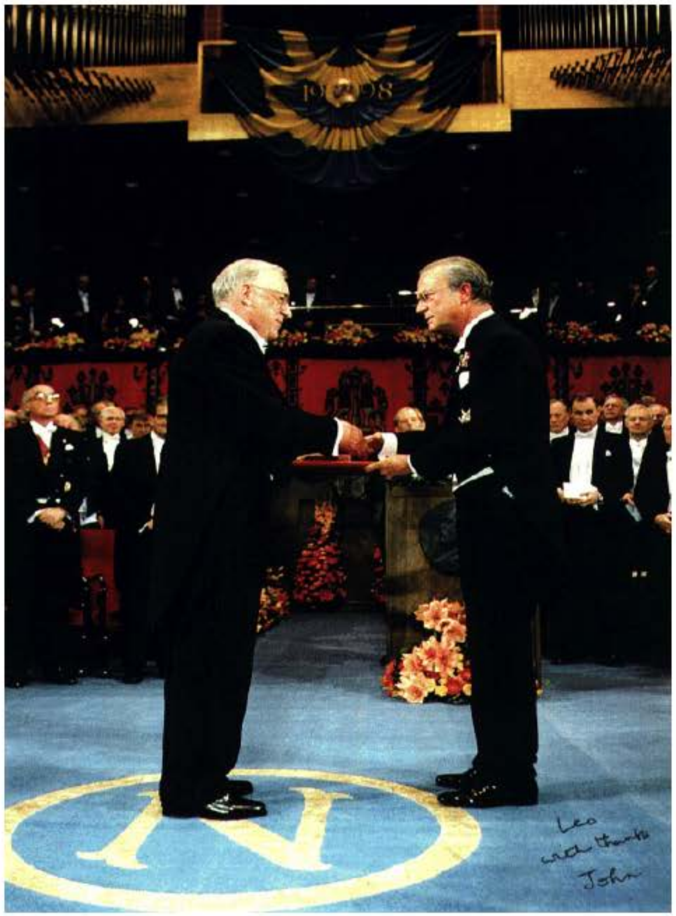
What Properties can be Calculated?

- Equilibrium structures
- Transition State structures
- Microwave, NMR spectra
- Reaction energies
- Reaction barriers
- Dissociation energies
- Charge distributions
- Reaction Rates
- Reaction Free Energies
- Circular Dichroism (optical, magnetic, vibrational)
- Spin-orbit couplings
- Full relativistic energies

What Properties can be Calculated?

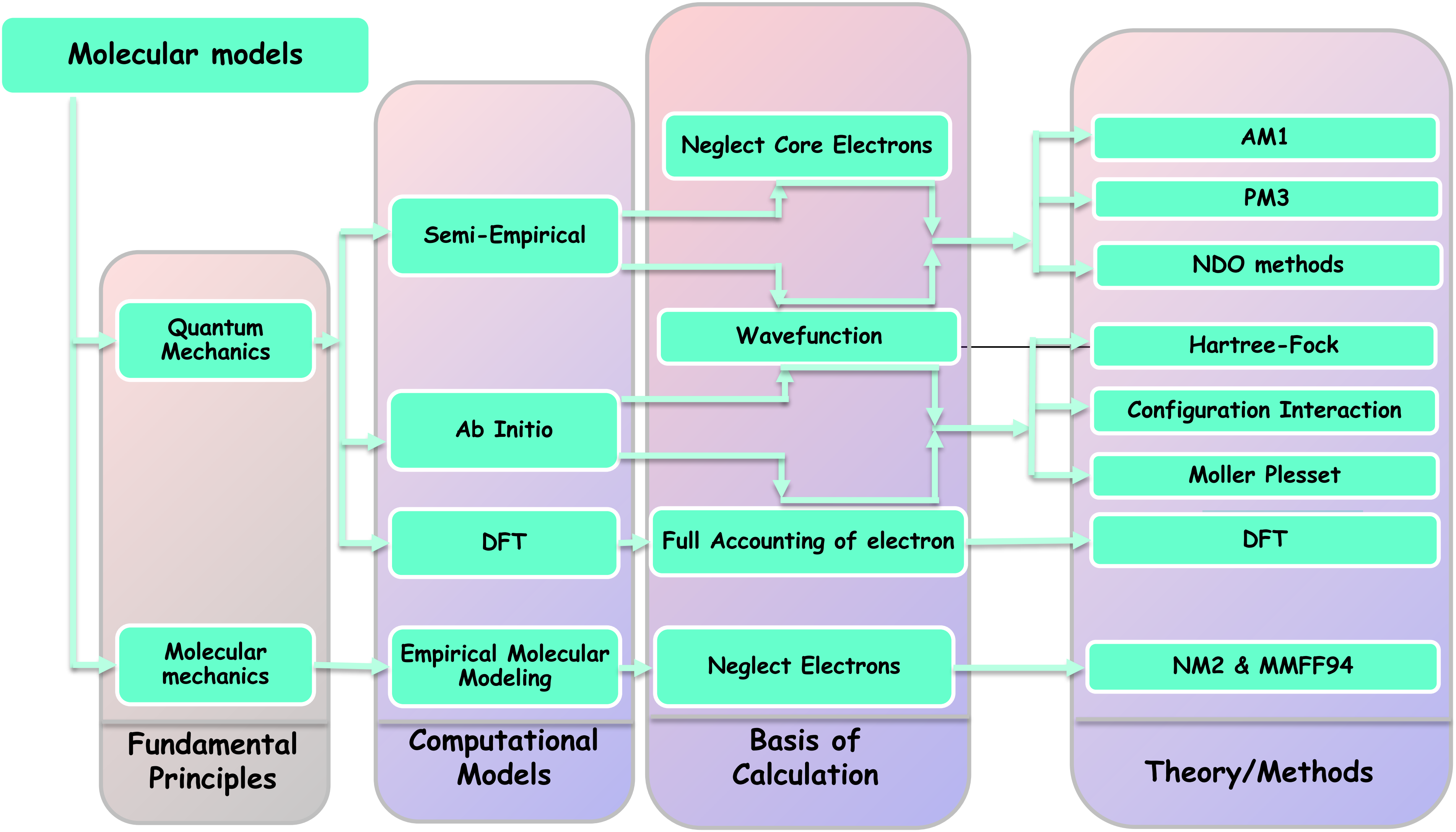
- Excited States (vertical)
- Solvent Effects
- pKa’s
- Density matrix methods/geminals
- Linear Scaling (ie of the methods with number of electrons/basis functions)
- Local correlation methods
- Accurate enzyme-substrate interactions
- Crystal structures
- Melting points
- Protein folding
- Full reaction dynamics
- Molecular dynamics
- Solvent dynamics
- Systematic improvements of DFT
- Excited States (adiabatic)…

Nobel recognition



The 1998 Nobel Prize in Chemistry was awarded to Walter Kohn "for his development of the density functional theory" and John Pople "for his development of computational methods in quantum chemistry".

Outline on Computational Chemistry



Computational Models

- A model is a system of equations, or computations used to determine the energetics of a molecule
- Different models use different approximations (or levels of theory) to produce results of varying levels of accuracy.
- There is a trade off between accuracy and computational time.
- There are two main types of models; those that use Schrödinger's equation (or simplifications of it) and those that do not.

Computational Models

- Schrödinger Equation can only be solved exactly for simple systems. Rigid Rotor, Harmonic Oscillator, Particle in a Box, Hydrogen Atom

- For more complex systems (i.e. many electron atoms/molecules) we need to make some simplifying assumptions/approximations and solve it numerically.

- However, it is still possible to get very accurate results (and also get very crummy results).

Quantum Mechanics

- Foundation is the Schrödinger Equation of quantum mechanics

$$\boxed{H\,\Psi = E\,\Psi}$$

- E is the energy of the system
- Ψ is the molecular wavefunction. Ψ has no simple physical meaning but Ψ^2 represents a probability distribution
- H is the Hamiltonian operator (a set of mathematical operations) describing the kinetic energy (T) and the potential energy (V) of the electrons and the nuclei

In principle we need to consider the electrons and nuclei in a molecule together, in practice, nuclei move much slower and we separate out electronic and nuclear motion (the Born-Oppenheimer approximation)

Quantum Mechanics

- The Schrödinger equation is the basis of quantum mechanics and gives a complete description of the electronic structure of a molecule.
  - Complex mathematical equation that completely describes the chemistry of a molecular system.
  - Not solvable for systems with many atoms.
  - Due to the difficulty of the equation computers are used in conjunction with simplifications and parameterizations to solve the equation.
- **Describes both the wave and particle behavior of electrons.**
  - The wavefunction is described by ψ while the particle behavior is represented by E.
  - In systems with more than one electron, the wavefunction is dependent on the position of the atoms; this makes it important to have an accurate geometric description of a system.

Quantum Mechanics

- Development of the Schrödinger equation from other fundamental laws of physics.

Kinetic Energy + Potential Energy = E

Classical
- Conservation of Energy
- Newton's Laws

$$\frac{1}{2}mv^2 + \frac{1}{2}kx^2 = E$$

Harmonic oscillator example.

$$F = ma = -kx$$

Quantum
- Conservation of Energy
- Schrodinger Equation

$$\frac{p^2}{2m} + \frac{1}{2}kx^2$$

The energy becomes the Hamiltonian operator

$$H\Psi = E\Psi$$

Wavefunction

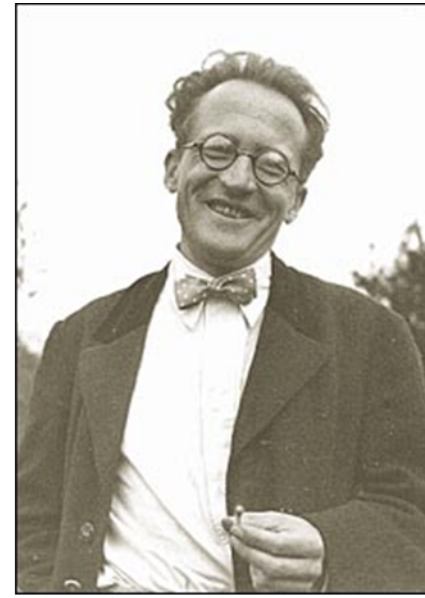
Energy "eigenvalue" for the system.

In making the transition to a wave equation, physical variables take the form of "operators".

$$p \rightarrow \frac{\hbar}{i}\frac{\partial}{\partial x} \qquad x \rightarrow x$$

$$H \rightarrow \frac{-\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + \frac{1}{2}kx^2$$

The form of the Hamiltonian operator for a quantum harmonic oscillator.



Erwin Schrödinger, 1927

http://hyperphysics.phy-astr.gsu.edu/hbase/quantum/schr.html

Don't worry……

Molecular Mechanics
(Classical Newtonian Physics)

- **Molecular mechanics is a computational method that computes the potential energy surface for a particular arrangement of atoms using potential functions that are derived using classical physics.**

- **Molecular mechanics ignores the electronic motions and calculate the potential energy of a system as a function of nuclear position only.**

- **Molecular mechanics methods are the basic for other methods, such as construction of homology models, molecular dynamics, crystallographic structure refinement and docking.**

Computational Models

Quantum mechanics

Semi Empirical

Uses experimental parameters and extensive simplification of Schrodinger’s equation.

Ab initio

Uses Schrodinger’s equation, but with approximations. No experimental data are included.

Density Functional Theory (DFT)

Considered an ab initio method, but different because the wavefunction is not used to describe a molecule, instead the electron density is used.

Molecular mechanics

Molecular Mechanics

It is based on classical physics to calculate force fields. It does not use Schrödinger's equation. Also called force field method. Atom treated as spheres, bonds and springs and electron are ignored.

Molecular Dynamics

It is a molecular mechanics program design to mimic the movement of atoms within a molecule. It can be carried out to a molecule to generate different conformation which on energy minimization, give arrange of stable conformation.

Computational Models

Parameters	Quantum mechanics		Molecular Mechanics
	Ab initio method	**Semi-empirical method**	
Molecular Size	Small	Medium	Large
Principle Calculation	Electronic Energy	Electronic Energy	Nuclear Energy
Time Required	Days	Hours	Minutes/Hours
Accuracy	High	Low	Low
Data Required	Computational	Experimental	Computational
Cost Affairs	High	Medium	Low

Semi Empirical

- used for calculations of molecules from 50-100 atoms
- borrows from both experimental and ab initio methods
  - attempts to simplify difficult mathematical equations in ab initio with SOME real data (empirical data) from the lab

- Like the *ab initio* methods, a Hamiltonian and wave function are used
  - much of the equation is approximated or eliminated
- Less accurate than *ab initio* methods but also much faster
- The equations are parameterized to reproduce specific results, usually the geometry and heat of formation, but these methods can be used to find other data.

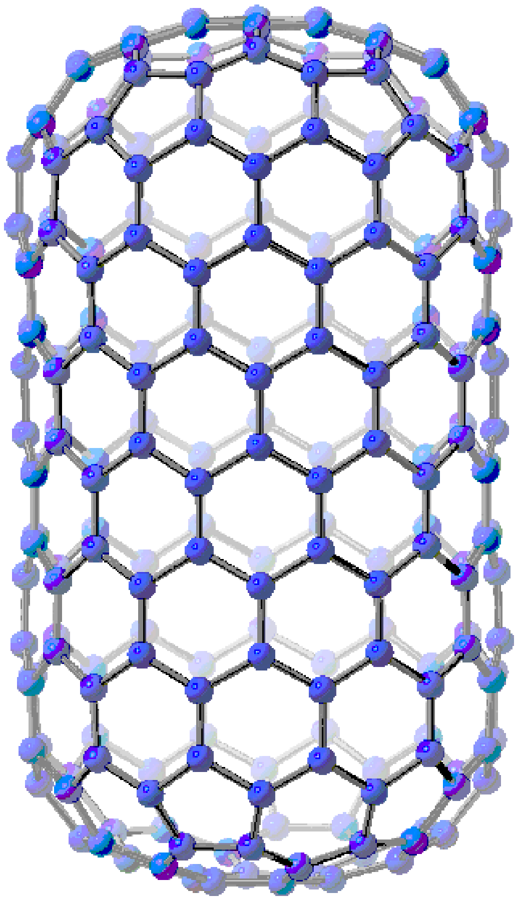
Ab Initio

- *Ab initio* translated from Latin means *"from first principles."* This refers to the fact that no experimental data is used and computations are based on quantum mechanics.
- *Ab initio* methods do not assume experimental parameters but instead try to calculate the molecular wavefunction directly using a variety of approximation techniques. These methods (like HF and MP2) can be very accurate, but can also be computationally expensive.
- Different Levels of *Ab Initio* Calculations
  - Hartree-Fock (HF)
    - The simplest *ab initio* calculation
    - The major disadvantage of HF calculations is that electron correlation is not taken into consideration.
    - Hartree-Fock methods (HF) with a mid-level basis set will often be used as a good starting point for more accurate calculations, or as a relatively fast way of getting qualitative data.
  - The Møller-Plesset Perturbation Theory (MP)
  - Density Functional Theory (DFT)
  - Configuration Interaction (CI)

Take into consideration electron correlation

Ab Initio

- Approximations used in some ab initio calculations
  - Central field approximation integrates the electron-electron repulsion term, giving an average effect instead of an explicit energy
  - Linear combination of atomic orbitals (LCAO): is used to describe the wave function and these functions are then combined into a determinant. This allows the equation to show that an electron was put in an orbital, but the electron cannot be specified.

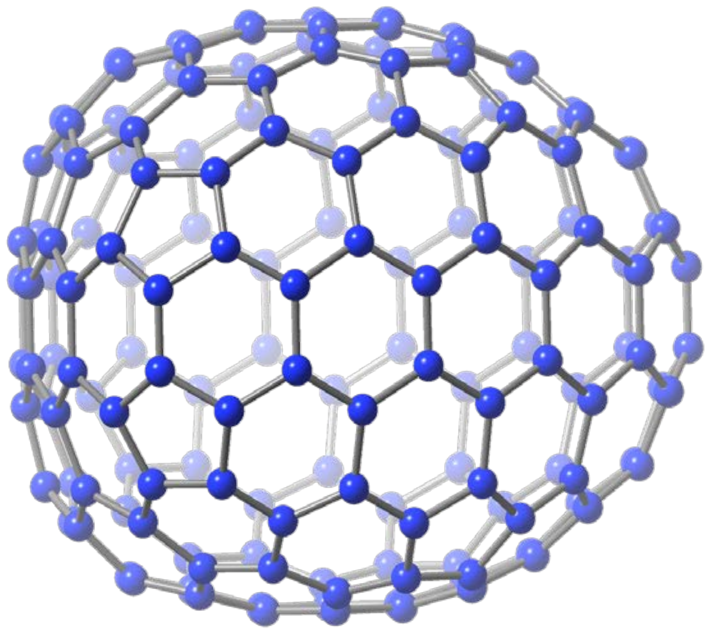


Simulated (12,0) zigzag carbon nanotube

Density Functional Theory

- Considered an *ab initio* method, but different from other *ab initio* methods because the wavefunction is not used to describe a molecule, instead the electron density is used.
- DFT is the most cost effective method to achieve a given level of quantitative accuracy. It includes electron correlation with less computational expense
- Three types of DFT calculations exist:
  – local density approximation (LDA) – fastest method, gives less accurate geometry, but provides good band structures
  – gradient corrected - gives more accurate geometries
  – hybrids (which are a combination of DFT and HF methods) - give more accurate geometries

Density Functional Theory



DFT: - 6139.46	HF: - 6132.89

% Difference: 0.11% (units: a.u.)

- These types of calculations are fast becoming the most relied upon calculations for nanotube and fullerene systems.
- DFT methods take less computational time than HF calculations and are considered more accurate
- This (15,0) short zigzag carbon nanotube was simulated with two different models: Hartree-Fock and DFT. The differences in energetics are shown in the table.

Molecular Mechanics

- Simplest type of calculation
  - Used when systems are very large and approaches that are more accurate become to costly (in time and memory)
- Does not use any quantum mechanics instead uses parameters derived from experimental or *ab initio* data
  - Uses information like bond stretching, bond bending, torsions, electrostatic interactions, van der Waals forces and hydrogen bonding to predict the energetics of a system
  - The energy associated with a certain type of bond is applied throughout the molecule. This leads to a great simplification of the equation

- assumes atoms to be like spheres at the end of a spring
- atoms obey classical mechanics
  - » Hooke's Law: F = -kx
  - » Newton's Second Law: F=ma
- used almost exclusively for molecules of 100 atoms or more
  - » proteins
  - » pharmaceuticals
  - » large biologicals

Computational Project

To Conduct a Computational Project these questions should be considered.

- ✓ What do you want to know?
- ✓ How accurate does the prediction need to be?
- ✓ How much time can be devoted to the problem?
- ✓ What approximations are being made?

The answers to these questions will determine the type of calculation, model and basis set to be used.

Basis Sets

- In chemistry a basis set is a group of mathematical functions used to describe the shape of the orbitals in a molecule, each basis set is a different group of constants used in the wavefunction of the Schrödinger equation.

- The accuracy of a calculation is dependent on both the model and the type of basis set applied to it.

- Once again there is a trade off between accuracy and time. Larger basis sets will describe the orbitals more accurately but take longer to solve.

- General expression for a basis function = $N * e^{(-\alpha * r)}$
  - where: N is the normalization constant, α is the orbital exponent, and r is the radius of the orbital in angstroms.

Basis Sets

- A basis set is a set of functions that restrict the electrons considered to specific regions of space
- Basis sets are basically approximations of atomic orbitals. ..the locations of electrons in the molecule (and thus parameters like geometry and energy) are estimated
- basis sets should also include polarization functions (account for distortions in orbital shapes), diffuse functions (necessary especially when you have a molecule with weakly bound electrons, for example: anions or transition states), and relativistic effects (for heavier atoms).
- Larger basis sets impose fewer restrictions, and so give better predictions
- However, larger basis sets are computationally more expensive

Examples of Basis Sets

- STO-3G basis set - simplest basis set, uses the minimal number of functions to describe each atom in the molecule
  - for nanotube systems this means hydrogen is described by one function (for the 1s orbital), while carbon is described by five functions (1s, 2s, 2px, 2py and 2pz).
- Split valence basis sets use two functions to describe different sizes of the same orbitals.
  - For example with a split valence basis set H would be described by two functions while C would be described by 10 functions.
  - 6-31G or 6-311G (which uses three functions for each orbital, a triple split valence set).
- Polarized basis sets - improve accuracy by allowing the shape of orbitals to change by adding orbitals beyond that which is necessary for an atom
  - 6-31G(d) (also known as the 6-31G*) - adds a d function to carbon atoms

Pople-style basis sets

- Named for Prof. John Pople who won the Nobel Prize in Chemistry for his work in quantum chemistry (1998).
- Notation: 6-31G

Use 6 primitives contracted to a single contracted-Gaussian to describe inner (core) electrons (1*s* in C)

Use 2 functions to describe valence orbitals (2s, 2p in C). One is a contracted-Gaussian composed of 3 primitives, the second is a single primitive.

6-311G   Use 3 functions to describe valence orbitals...

6-31G*   Add functions of ang. momentum type 1 greater than occupied in bonding atoms (For N_2 we'd add a d)

6-31G(d)   Same as 6-31G* for 2nd and 3rd row atoms

Common Types of Basis Sets

- Pople split-valence basis sets
  - More basis functions lead to better description of orbital wavefunction
  - Described in the form x-yz+G(p,d)
  - Polarization functions (*p* and *d* type functions) are added to describe *s* and *p* distortion
  - Diffuse functions (+) are added for delocalized charges
  - Carbon

	1s **core orbital** x gaussians	$2s, 2p_x, 2p_y, 2p_z$ **inner valence orbital** y gaussians	$2s, 2p_x, 2p_y, 2p_z$ **outer valence orbital** z gaussians	**other** +, p, d	**total primitives**	**total basis functions**
3-21G	3	2	1		16	9
6-31G	6	3	1		22	9
6-31G(d)	6	3	1	6	28	15
6-31+G	6	3	1	4	26	13
6-311G	6	3	1+1		26	13

Accuracy

The accuracy of the computed properties is sensitive to the quality of the basis set. Consider the bond length and dissociation energy of the hydrogen fluoride molecule:

Basis set	Bond Length (Å)	D_0 (kJ/mol)
6-31G(*d*)	0.9337	491
6-31G(*d*, *p*)	0.9213	523
6-31+G(*d*)	0.9408	515
6-311G(*d*)	0.9175	484
6-311+G(*d*, *p*)	0.9166	551
Expt.	0.917	566

ZPVE = 25 kJ/mol MP2/6-311+G(*d*, *p*)

Tools

Over 100 commercially available software packages

CHARMM
(Molecular modelling & Molecular Dynamics)

The Nobel Prize in Chemistry 2013

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Martin Karplus

Photo: Keilana via Wikimedia Commons
Michael Levitt

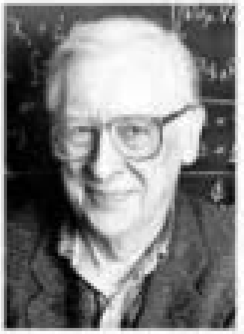
Photo: Wikimedia Commons
Arieh Warshel

www.charmm.org

Gaussian
(Quantum mechanics)

The Nobel Prize in Chemistry 1998

Walter Kohn
Prize share: 1/2



John A. Pople
Prize share: 1/2

www.gaussian.com

Widely used tools for computational lab

Package	License	Design to	Use
Avogadro	Free	Teaching Visualization, Input Generation for other Packages	Molecular modeling, bioinformatics, etc.
accelrys MATERIALS STUDIO	Academic, Commercial	Bioinformatics, cheminformatics, atomic elements rebuilding, and quantum mechanics.	Utilized in cutting-edge exploration of different materials, like polymers, carbon nanotubes, metals, ceramics, etc., by colleges, research focuses, and innovative organizations.
GROMACS FAST. FLEXIBLE. FREE.	Free	Intended for reproductions of proteins, lipids, and nucleic acids.	One of the quickest and most famous programming bundles accessible and can run on focal handling units (CPUs) and illustrations preparing units (GPU)
NAMD Scalable Molecular Dynamics	Free	A vast number of atoms	NAMD is a parallel molecular dynamics code designed for high-performance simulation of large biomolecular systems.
Expanding the limits of computational chemistry	Commercial	Utilizes fundamental laws of quantum mechanics	Used to predict energies, molecular structures, spectroscopic data (NMR, IR, UV, etc)

Widely used tools for computational lab

- There exist a large number of software packages
  - MOLPRO, GAMESS, COLUMBUS, NWCHEM, MOLFDIR, ACESII, GAUSSIAN, Hyperchem, WebMO
  - The different programs have various advantages and capabilities.
- Gaussview is a GUI that interfaces with Gaussian for aiding in building molecules and viewing output.
- GAMESS
  - Text-only interface, even more bare than Gaussian
  - Free (http://www.msg.ameslab.gov/GAMESS/)
  - Well-known and used by researchers
- Titan
  - Easy to use GUI
  - Not as flexible as other programs

Advantages of Material Studio Software

World's most comprehensive Environment for Computational Materials Science

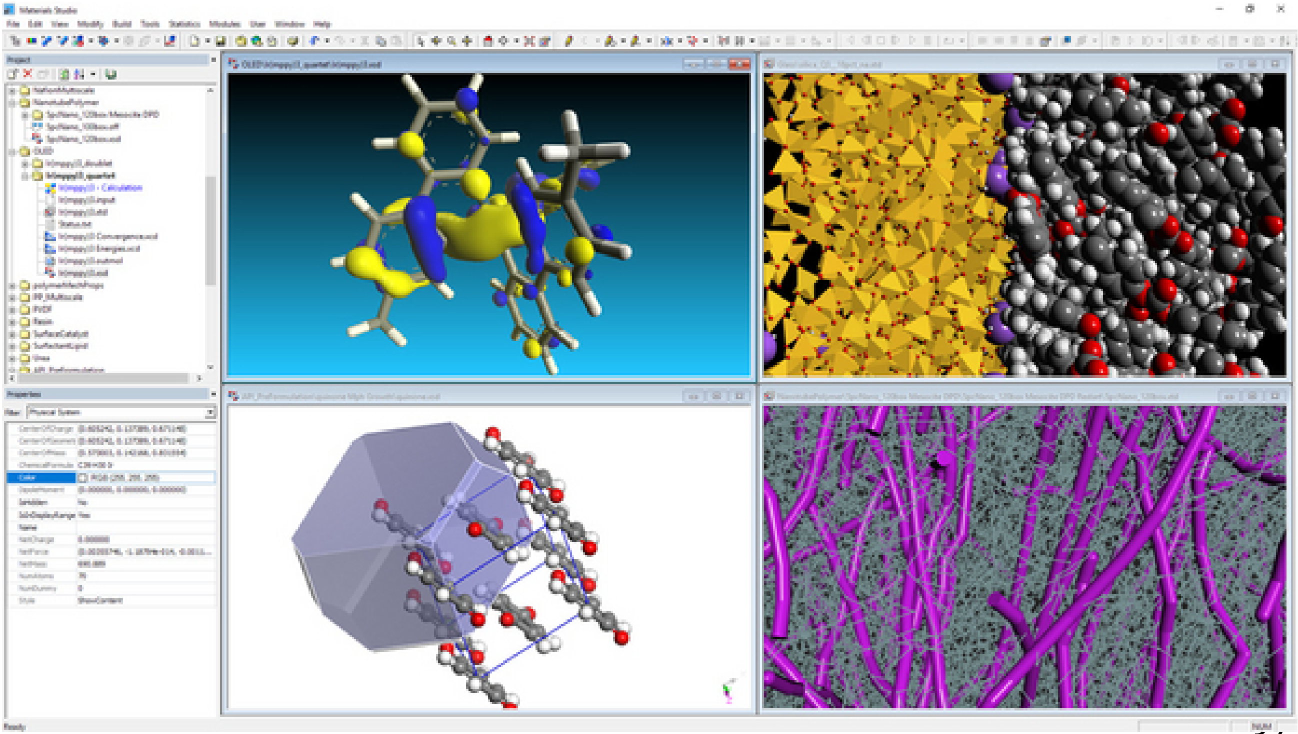
- **Visual model building**
- **Computation**
- **Results post processing**

Most complete set of computational methods

- **Quantum mechanics**
- **Classical simulation**
- **Mesoscale modeling**

Widest range of materials

- **Polymers**
- **Catalysts**
- **Crystals**
- **Nano-structured materials**
- **And more . . ...**



DRUG DISCOVERY

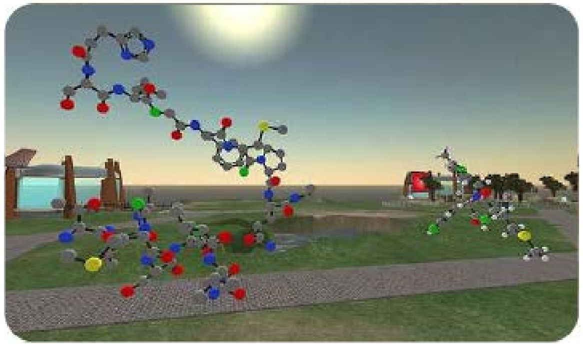
- ❖ Drug discovery take years to decade for discovering a new drug.
- ❖ Effort to cut down the research timeline and cost by reducing wet-lab experiment use computer modelling.
- ❖ Drugs interact with their receptors in a highly specific and complementary manner.

Drug targets

- ❖ Enzyme – inhibitors
- ❖ Receptors - agonists or antagonists
- ❖ Ion channel – blockers
- ❖ Transporter –inhibitors
- ❖ DNA - blockers

What is CADD

- Computer Aided Drug Design (CADD) is the chemistry whose major goals are to create efficient mathematical approximations and computer programs that calculate the properties of future drug molecules and thus helping in the process of drug design and discovery.

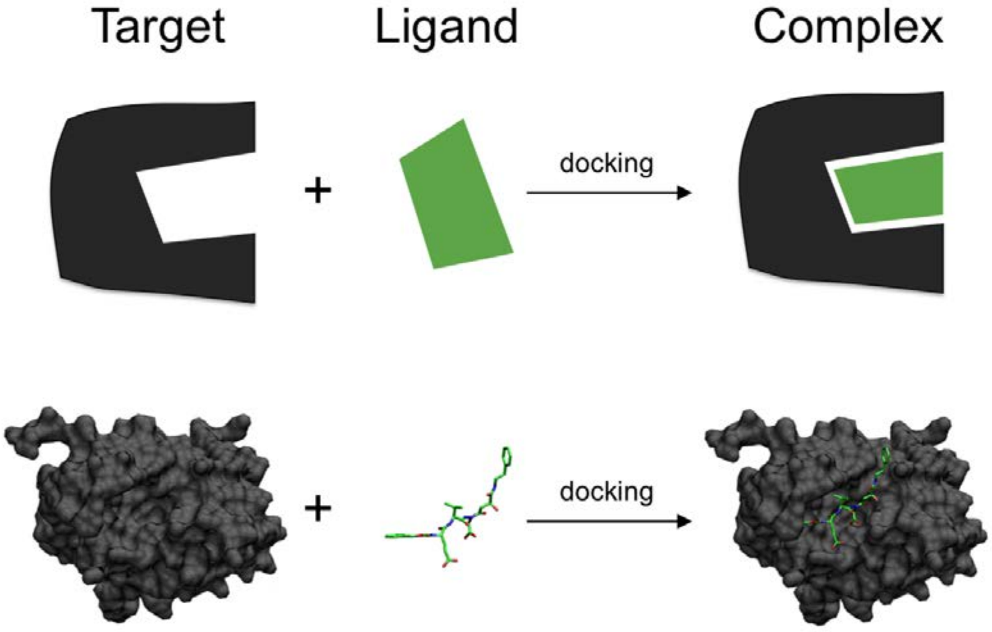


- Drug Discovery today are facing a serious challenge because of the increased cost and enormous amount of time taken to discover a new drug, and also because of rigorous competition amongst different pharmaceutical companies.

Molecular docking

Molecular docking is the study of how two or more molecular structures (e.g., drug and enzyme or protein) fit together .

docking is a molecular modeling technique that is used to predict how a protein (enzyme) interacts with small molecules (ligands).



WHAT IS DOCKING?

Docking is a structure-based technique which attempts to find the "best"match, between two molecules

IMPORTANCE OF DOCKING

- It is the key to rational drug design.
- The results of docking can be used to find inhibitors for specific target proteins and thus to design new drugs.
- It is gaining importance as the number of proteins whose structure is known increases.
- Identification of the ligand’s Correct binding geometry (pose) in the binding site.
- Signal transduction

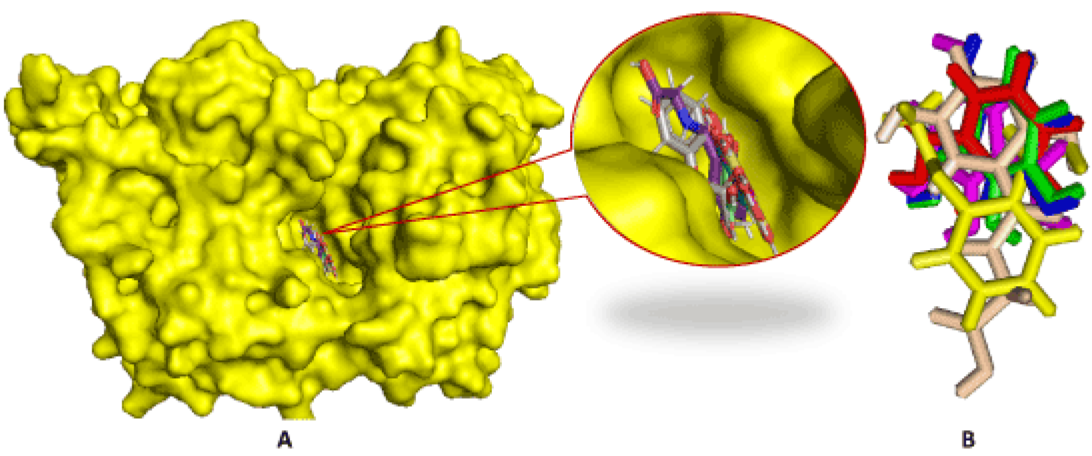
DOCKING GLOSSARY

- **Receptor or host or lock:** the "receiving" molecule, most commonly a protein or other biopolymer.
- **Ligand or guest or key:** the complementary partner molecule which binds to the receptor. Ligands are most often small molecules but could also be another biopolymer.
- **Docking:** computational simulation of a candidate ligand binding to a receptor.
- **Binding mode:** the orientation of the ligand relative to the receptor as well as the conformation of the ligand and receptor when bound to each other.
- **Pose :** a candidate binding mode.
- **Scoring :** the process of evaluating a particular pose by counting the number of favorable intermolecular interactions such as hydrogen bonds and hydrophobic contacts.
- **Ranking** : the process of classifying which ligands are most likely to interact favorably to a particular receptor based on the predicted free-energy of binding

Molecular docking

Molecular docking is one of the most frequently used methods in structure-based drug design, due to its ability to predict the binding-conformation of small molecule ligands to the appropriate target binding site.

Characterization of the binding behavior plays an important role in rational design of drugs as well as to elucidate fundamental biochemical processes.



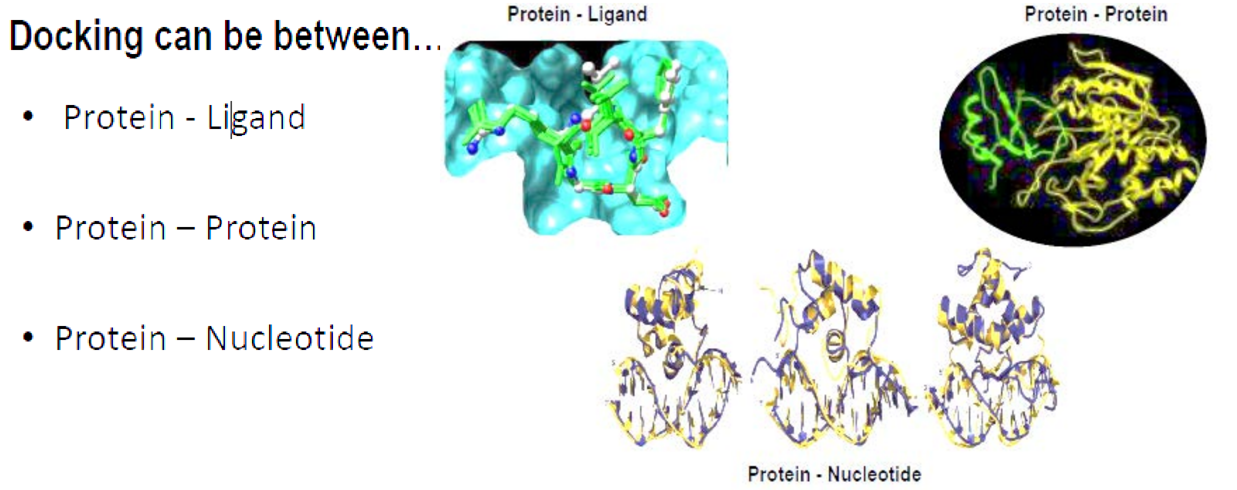
Types of docking

Rigid Docking (Lock and Key)

In rigid docking, the internal geometry of both the receptor and ligand are treated as rigid. In the rigid docking molecules are rigid, in 3D space of one of the molecule which brings it to an optimal fit with the other molecules in terms of a scoring function.

Flexible Docking (Induced fit)

In flexible docking molecules are flexible. An enumeration on the rotations of one of the molecules (usually smaller one) is performed. Every rotation the energy is calculated; later the most optimum pose is selected.



TYPES OF INTERACTIONS

Electrostatic forces - Forces with electrostatic origin are due to the charges residing in the matter.

Electrodynamics forces - The most widely known is probably the van der Waals interaction.

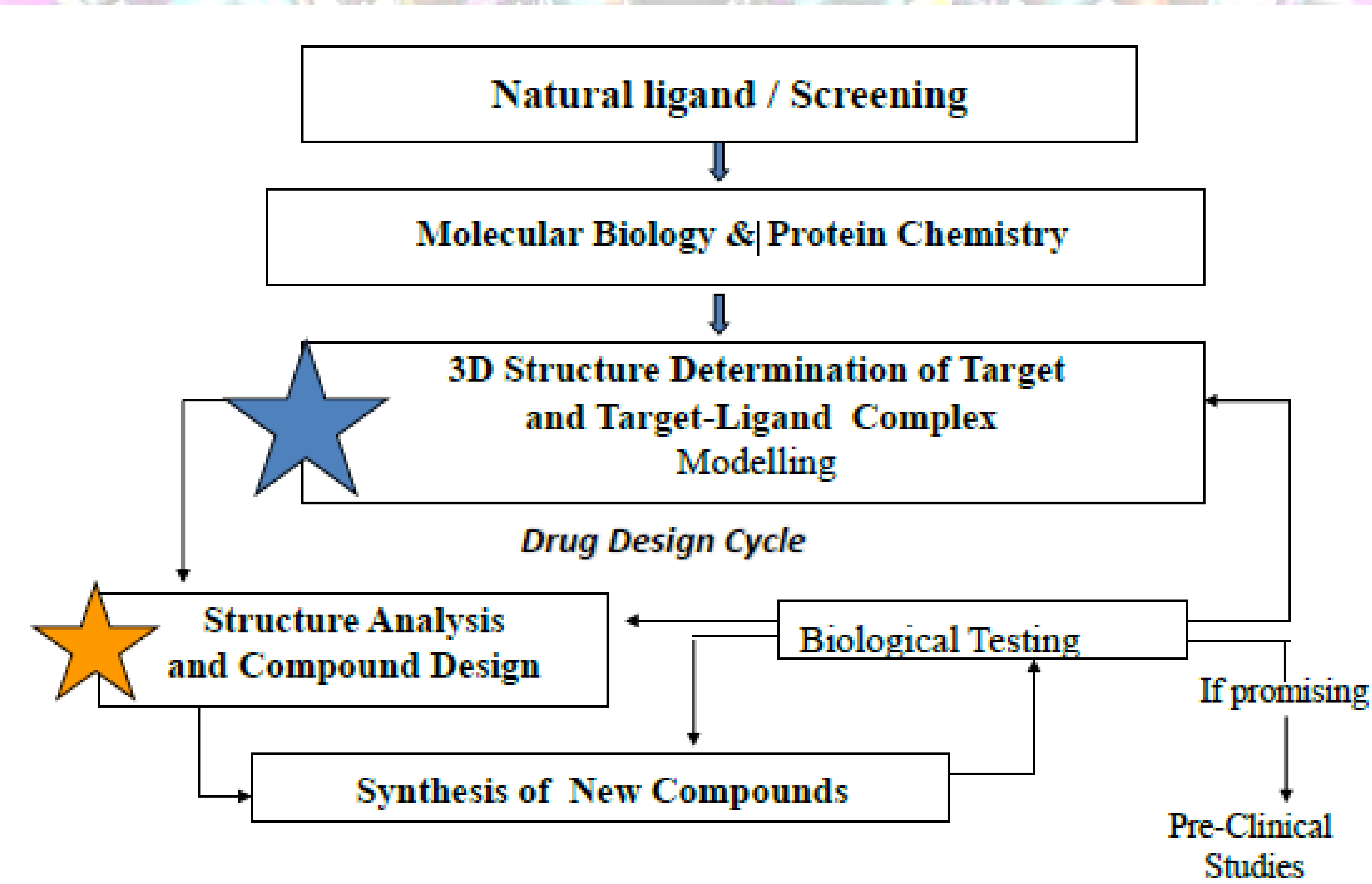
Steric forces - These are caused by entropy. For example, in cases where entropy is limited, there may be forces to minimize the free energy of the system.

Solvent-related forces – These are due to the structural changes of the solvent. These structural changes are generated, when ions, colloids, proteins etc, are added into the structure of solvent. The most commonly are Hydrogen bond and hydrophobic interactions

Key stages in docking

- **Target/Receptor selection and preparation**
- **Ligand selection and preparation**
- **Docking**
- **Evaluating docking results**

Modern Drug Discovery Process



Application of Molecular Docking in Modern Drug Discovery

Case Study-

Identification of Inhibitors for Simultaneous Inhibition of Anti-Coagulation and Anti-Infamatory Activities of Snake venom Phospholipase A2. (1994)

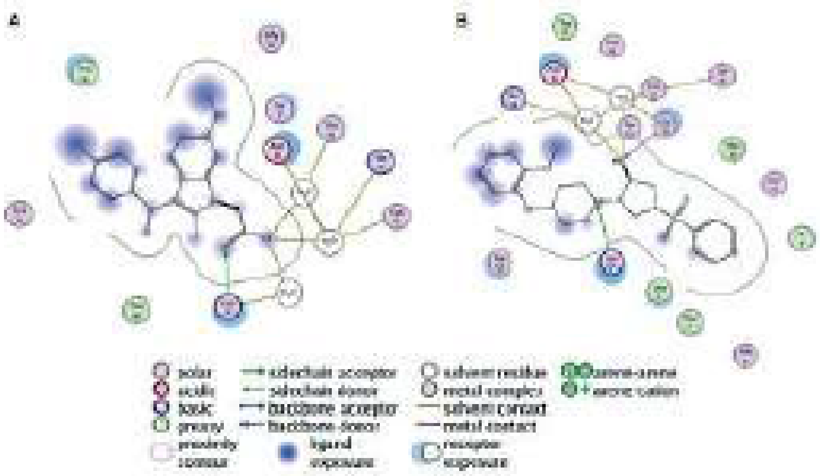
169000 compounds

Pharmacophore Based-screening

300 compounds

Molecular docking

32 promising compounds



Docking Tools:

- PyMol
- PyRx
- Discover Studio
- Open Babel
- FlexX (1996)
- Hammerhead (1996)
- Surflex (2003)
- SLIDE (2002)
- ICM (1994)
- MCDock (1999)
- GOLD – University of Cambridge ,UK
- AUTODOCK - Scripps Research Institute,USA (autodock.scripps.edu/)
- GemDock (gemdock.life.nctu.edu.tw/dock/)
- Hex Protein Docking - (hex.loria.fr/)
- GRAMM (Global Range Molecular Matching) (www.bioinformatics.ku.edu/files/vakser/gramm/)
- FRED (2002)
- Glide (2004)
- Yucca (2005)
- vLife(Biopredicta)

Thank You